



DIGITAL TRUST FUTURES
FOUNDATION

DIGITAL SOVEREIGNTY IN PRACTICE

Real Digital Sovereignty Starts With Control

Five pillars every country must understand

Digital Sovereignty Series | Article 1 of 6

Why this topic matters now

Countries digitising identity, payments, health, welfare and data exchange systems are making sovereignty decisions whether or not those decisions are named. The first article in this series clarifies what digital sovereignty actually means, and why local hosting, vendor restrictions or open-source adoption alone can create false confidence if mistaken for real control.

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Digital sovereignty is demonstrated control

Digital sovereignty is often used as a political or procurement phrase, but its practical meaning is much stricter: a country must be able to control, operate, assess, secure and adapt its own critical digital systems and the data that flows through them. It is not achieved by a single law, a data centre, a local vendor preference, or the use of open-source software alone.

This opening article frames the series' core argument. Sovereignty is a system of interlocking capabilities: architectural, data, operational, governance and security sovereignty. If one pillar is missing, the country may appear sovereign but lack the substance of control.

Key points for policymakers and DPI leaders

Digital sovereignty, at its simplest, is a country's demonstrated ability to control the digital systems and data on which its people, institutions and economy increasingly depend. The word **demonstrated** matters. Sovereignty cannot be inferred from a press release, a procurement clause or a cloud region name; it must be visible in architecture, operations, governance, security assurance, and the country's ability to continue functioning under stress.

- Digital sovereignty is demonstrated control, not a declaration or branding claim.
- A local data centre can support residency, but it does not automatically create legal or technical control.
- Open-source Digital Public Goods are a powerful starting point, but they solve only part of the sovereignty problem.
- Sovereignty should be treated as an operating discipline that has to be sustained, tested and funded over time.

The real fear is not foreign technology itself. International technology can be useful, efficient and in many cases necessary. The risk is **dependency without recourse**: a position in which another company, government or ecosystem can shape, restrict, expose or disrupt a country's critical systems, while the country lacks the practical means to prevent, detect or recover from such disruptions.

The sovereignty question for leaders is not, "Where is the server?" The stronger question is: "If the vendor, cloud provider, donor or integration partner became unavailable or legally constrained tomorrow, could we keep this system running, protect the data, and know quickly if something had gone wrong?"

The five pillars of digital sovereignty

Digital sovereignty is the cumulative product of five interlocking conditions. A country may be strong on one and weak on the rest, but it should not confuse that partial progress with full control. The table below sets out the practical pillar framing that underpins every subsequent article in the series.

Pillar	What it means in plain terms	Failure mode if missing
1. Architectural sovereignty	Critical systems can be operated, maintained and modified by domestic technical capability without being permanently locked to a single foreign vendor.	The country cannot fix, extend or replace its own identity or payment system without first calling a foreign supplier.
2. Data sovereignty	Citizens' data stays under the legal and technical authority of the state that collected it, enforced technically, not just promised in policy.	Data sits in-country, but a foreign government can still compel access to it through that provider's home-country law.
3. Operational sovereignty	The country can keep critical systems running, including in degraded conditions, with a trained domestic workforce and tested manual fallback procedures.	A foreign vendor's outage, dispute or sanctions exposure becomes a national service outage.
4. Governance sovereignty	Decisions about architecture, security standards and access policy are made by accountable domestic institutions, not effectively outsourced to a vendor or donor.	Strategic technology decisions are made in a foreign vendor's roadmap meeting, not in a national ministry.
5. Security sovereignty	The state can independently assess the security of its own systems without relying on the cooperation of the parties that built them.	The only people checking whether the system is secure are the ones who profit from claiming it is.

The honest test of sovereignty

A country's digital infrastructure is genuinely sovereign when it can answer yes, without hesitation, to one question: "If every vendor we work with went dark tomorrow, could we keep this system running, keep citizens' data safe, and know within 24 hours if something had gone wrong?"

Most countries currently deploying Digital Public Infrastructure cannot yet answer yes.

The five-pillar model is deliberately practical. It is intended to move digital sovereignty away from slogans, declarations and procurement language, and toward a clear test of whether a country has real control over the systems on which its citizens, economy and public services increasingly depend.

1. Architectural sovereignty

This pillar asks whether the country can inspect, modify, integrate and operate the system without being permanently dependent on a single foreign vendor or external technical party. This does not mean rejecting foreign technology or external implementation support. It means ensuring that the core architecture, source code, standards, interfaces, documentation and operational knowledge are sufficiently open, transferable and understood by domestic or trusted regional teams. A country lacks architectural sovereignty when it cannot change its own identity, payment, data exchange or welfare system without waiting for a vendor roadmap, foreign approval, or specialist support that it cannot replace.

2. Data sovereignty

This pillar asks whether citizens' data remains under domestic legal and technical authority, not merely whether it is stored inside the country's borders. This is the pillar most often confused with data

localisation. Data sovereignty requires clarity over who controls the data, who can access it, who holds the encryption keys, what laws apply, and whether foreign legal or corporate structures can override national policy. A country may have local hosting and still lack data sovereignty if a foreign-parented provider, a foreign court, or an external operator can compel or facilitate access to sensitive citizen data without meaningful domestic control.

3. Operational sovereignty

This pillar asks whether the system can continue to run when vendors, networks, funding conditions, geopolitical pressures or political relationships become adverse. This is the practical test of sovereignty's continuity. It requires trained domestic teams, tested fallback procedures, incident-response capability, backup and recovery arrangements, service continuity planning, and the ability to operate under degraded conditions. A country that can launch a system but cannot keep it running without the constant presence of an external implementer has achieved deployment, not sovereignty.

4. Governance sovereignty

This pillar asks who actually makes the critical decisions about architecture, data access, security standards, procurement, risk acceptance, incident disclosure and future system evolution. Formal ownership by a ministry or agency is not enough if the real decisions are shaped elsewhere, by donor conditions, vendor defaults, opaque technical committees or commercial platform limitations. Governance sovereignty means that accountable domestic institutions have the authority, competence, evidence and independence to make informed decisions in the national interest, and to challenge external parties when necessary.

5. Security sovereignty

This pillar asks whether independent assurance exists beyond the parties that built, fund, host or operate the system. This pillar is essential because a country cannot simply rely on vendor claims, implementation reports or compliance checklists to know whether a critical national system is secure. Security sovereignty requires independent testing, code review, supply chain assurance, vulnerability management, monitoring, audit rights, incident readiness and maturity assessment. Without this pillar, the country may believe it controls the system, but it has no independent way to verify whether that control is technically real, resilient and defensible under attack.

The five common mistakes

Digital sovereignty is often weakened not by a lack of ambition but by a series of recurring misunderstandings that lead governments to believe they have more control than they actually do. These mistakes matter because they can shape national procurement decisions, cloud strategies, DPI deployment choices and public communication.

A country may announce sovereignty, build local hosting arrangements, adopt open-source tools or restrict foreign vendors, and still remain dependent at the legal, technical, operational or security level. The danger is that these measures can create the appearance of control while leaving the real levers of control elsewhere. For policymakers, the first step is therefore to separate useful sovereignty-enabling measures from the mistaken belief that any one of them is sovereignty by itself.

Mistake 1: A local data centre means the data is sovereign

Local hosting can be useful. It can reduce latency, satisfy residency rules, support local investment and make some operational controls easier. But it is not the same as sovereignty if the company operating the infrastructure remains subject to another jurisdiction's law and if the encryption keys, administrative access or support channels remain outside domestic control.

Mistake 2: Restricting foreign vendors creates sovereignty

Vendor restriction addresses one layer of procurement. It does not automatically resolve foreign dependence in cloud infrastructure, network equipment, undersea cable capacity, managed services, developer tooling or software updates. A domestic prime contractor can still deliver a foreign-controlled digital stack.

Mistake 3: Open source removes dependency

Open-source Digital Public Goods reduce proprietary lock-in and are often the right starting point for sovereign DPI. But open-source code still runs on infrastructure, processes personal data, uses dependencies, receives updates and relies on development communities. Those layers require separate controls.

Mistake 4: Sovereignty requires digital isolation

Some governments respond to dependency risk by trying to wall themselves off from global technology. That can reduce access to innovation, increase costs and weaken security if the domestic alternative is under-resourced. The better objective is verified control, not isolation.

Mistake 5: Sovereignty is a one-time achievement

Sovereignty erodes when new vendors are added, integrations expand, AI components are introduced, keys are delegated, staff leave, or security assurance becomes stale. It therefore has to be maintained as a living discipline.

The common thread across all five mistakes is that sovereignty is not created by a single location, vendor rule, software licence, policy statement or launch milestone. It is created by the continuous ability to govern, operate, secure, verify and adapt the systems that carry national data and public trust. Real digital sovereignty begins when a country can distinguish between symbolic control and control that remains true under legal pressure, technical failure, vendor dependency and active attack.

Series bridge

The next article turns from definition to consequence. It asks what a country risks when it mistakes the appearance of sovereignty for the reality of control, especially in population-scale identity, payment, welfare, health and data-exchange systems.

About the author

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Godfrey Kutumela is a globally recognised cybersecurity practitioner with over 26 years of experience shaping secure digital ecosystems across fintech, identity and public infrastructure environments. He operates at the intersection of Digital Public Infrastructure (DPI), Digital Public Goods (DPGs) and national-scale platforms, bringing deep expertise in securing systems where the impact of failure is measured at the population level rather than the enterprise level.

- Trusted advisor to governments, foundations and ecosystem builders delivering secure-by-design DPI across Africa, Latin America and global markets.
- Core contributor to large-scale platforms including Mowali (mobile money interoperability infrastructure), Mojaloop (interoperable faster payments), MOSIP (digital identity) and Tazama (fraud systems).
- Designs and embeds independent, risk-based security practices across payment systems, identity frameworks and cross-border financial infrastructure.
- Works across policy, architecture and implementation layers, translating global standards into deployable security controls.
- Contributor to the Digital Public Goods Alliance (DPGA) Privacy Expert Group, shaping global privacy and security indicators.
- Active in open-source security, DevSecOps integration and supply chain assurance for reusable digital platforms.
- Delivered independent security reviews, penetration testing and architecture governance across critical DPI implementations.

His leadership philosophy in the DPI ecosystem is clear: security forms the foundation upon which safety and trust are built. Safety is what citizens experience in practice, and trust is the legitimacy that sustains adoption; once lost, it is difficult to restore.

About Digital Trust Futures Foundation

Digital Trust Futures Foundation is an independent, non-profit public-benefit organisation strengthening the safety, security, rights, inclusion and trustworthiness of digital public infrastructure (DPI), digital public goods (DPGs), open-source technology and AI-enabled public services. Its work is grounded in the recognition that DPI has become critical national infrastructure operating in a persistent threat environment, requiring independent, ecosystem-level assurance that goes beyond traditional approaches.

The Foundation operates outside vendor, implementer and platform incentives. It combines evidence-based cybersecurity, safety and AI-governance assurance with human-rights and child-protection expertise, and releases its tools, research and frameworks as reusable public goods rather than proprietary products — so that the value of a single engagement extends to every government, maintainer and community facing the same risk.

Digital Trust Futures Foundation NPC is registered as a non-profit company in South Africa and works across Africa and the wider Global South, with emphasis on countries and communities adopting DPI under capacity and assurance constraints.

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